



# **Department of Electrical and Information Engineering**

**University of Ruhuna**

## **Mini Project Proposal**

**Artificial Intelligence  
EC6301**

## **Student Marks Prediction System Using Machine Learning**

**G20**

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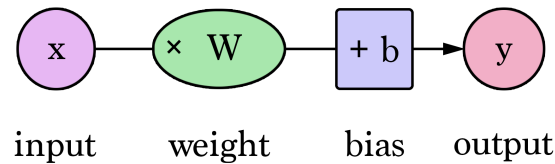
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## 1. Problem Statement

Usually student performance is evaluated only after final examinations, which limits the ability of educators to take correct actions in time.

Educational institutions generate a large amount of data related to student performance, including attendance records, assignment marks, quiz scores, and mid-term examination results. Also Students have various learning patterns However, this data is rarely used for predictive analysis. Manual analysis of such data is time-consuming, inefficient, and prone to human error. Furthermore, conventional statistical methods are often not enough in capturing complex and nonlinear relationships among multiple factors affecting student performance.

With ANN, it is possible to identify are there any patterns between students results and learning how far those factors affect(weights) to the final result.



**Figure 1: Neurone**

However to create those Artificial Neural Network a demanding processing power is needed. Also it is difficult to interpret how the model makes decisions internally. That leads us towards to question the reliability of ANN model.

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## 2. Objectives

The primary objective of this mini project is to design and develop a machine learning-based system capable of predicting student examination marks using historical academic data. By leveraging Artificial Intelligence techniques, the project seeks to support educators in making informed academic decisions.

One of the key objectives is to analyze various factors that influence student academic performance. These factors include attendance percentage, assignment marks, quiz scores, mid-term examination results, and study habits. Understanding the impact of these factors is essential for developing an accurate and reliable prediction model.

Another important objective is to implement a prediction model using Artificial Neural Networks. Neural networks are well suited for handling complex and nonlinear relationships within data. The project aims to demonstrate the effectiveness of neural networks in predicting student marks compared to traditional statistical methods.

The project also aims to evaluate the performance of the developed model using appropriate evaluation metrics such as Mean Squared Error and prediction accuracy. This evaluation will help determine the reliability and practical applicability of the system.

Overall, the project aims to contribute to the application of Artificial Intelligence in the education sector by developing a practical and data-driven student performance prediction system.

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### 3. Literature Review

Several research studies have explored the application of machine learning techniques for predicting student academic performance.

Recent studies have shown that machine learning techniques can effectively predict student academic performance [1, 2]. Neural network-based approaches have demonstrated higher accuracy compared to traditional statistical methods [3, 4]. Educational data mining has become an important research area in recent years [5, 6].

Early research primarily utilized statistical approaches such as linear regression to examine the relationship between student attributes and academic outcomes. Although these approaches provided baseline predictions, they were limited in handling complex data patterns.

Decision tree and ensemble learning techniques such as Random Forest have also been applied in this domain. These models offer better interpretability and help identify key factors affecting student performance. However, their predictive performance can vary depending on dataset size and feature selection.

Some studies have incorporated deep learning approaches to analyze behavioral and academic data. While these approaches achieve high accuracy, they often require large datasets and significant computational resources. Despite these advancements, many existing studies focus on limited datasets or lack practical implementation.

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## 4. Methodology

The proposed Student Marks Prediction System follows a structured machine learning-based methodology. Initially, a dataset containing student academic records will be collected from publicly available sources or generated synthetically for experimental purposes. The dataset will include features such as attendance percentage, assignment marks, quiz scores, mid-term examination results, and study hours.

The collected data will undergo preprocessing to improve its quality and suitability for machine learning. This includes handling missing values, removing inconsistencies, and normalizing numerical features. Data preprocessing is a critical step to ensure accurate and reliable model performance.

An Artificial Neural Network will be used as the primary machine learning model. The dataset will be divided into training and testing subsets to evaluate the generalization capability of the model. The neural network will be trained using the training dataset and optimized by adjusting model parameters.

After training, the model will be evaluated using the testing dataset. Performance metrics such as Mean Squared Error and prediction accuracy will be used to assess model effectiveness. The trained model will then be used to predict final examination marks for new student data.

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## 5. Dataset / Data Collection

The dataset used in this project will consist of student academic records obtained from publicly available educational datasets or synthetically generated data. The dataset will include numerical and categorical attributes related to student performance. Key features include attendance percentage, assignment marks, quiz scores, mid-term examination results, and study hours.

To ensure data quality, preprocessing techniques such as normalization, handling missing values, and data validation will be applied. Only anonymized data will be used to maintain ethical standards and protect student privacy. The dataset will be structured in a tabular format suitable for machine learning analysis.

Data diversity will be ensured by including records from students with varying academic backgrounds and performance levels. Proper data collection and preparation are essential to ensure that the machine learning model produces accurate and reliable predictions.

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## 6. Expected Outcomes

The expected outcome of this project is the development of a functional Student Marks Prediction System based on machine learning techniques. The system will be capable of predicting student examination marks with acceptable accuracy.

A trained Artificial Neural Network model will be produced as a key outcome. The model is expected to demonstrate improved prediction performance compared to traditional evaluation methods. Performance evaluation results, including accuracy metrics and error analysis, will be generated to validate system effectiveness.

The system will also provide valuable insights into factors affecting student performance. Educators can use these insights to identify students at risk of poor academic performance and take timely corrective actions. Overall, the project aims to demonstrate the practical application of Artificial Intelligence in education.

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## 7. Percentage of AI Involvement

Artificial Intelligence plays a central role in this project, accounting for approximately 80% of the overall system. Machine learning techniques are applied during data preprocessing, feature analysis, model training, and prediction. The Artificial Neural Network serves as the core component, enabling intelligent learning and accurate prediction of student academic performance.

Non-AI components such as documentation and result visualization form a smaller portion of the system. Therefore, the project clearly satisfies and exceeds the required AI involvement percentage.

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